Adverse Childhood Experiences as an Early Risk Factor for Alzheimer's Disease

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Introduction and Lit Review

Alzheimer's Disease (AD), a chronic, progressive disorder characterized by structural and functional changes in the brain, has been growing to substantial amounts in recent years. According to Abouelmagd et al., PhD Researcher in the Faculty of Medicine at Mansoura University, approximately 47 million people were diagnosed with neurodegenerative disorders, and this number is expected to triple by 2050, affecting nearly 70% of the population. (Abouelmagd et al., 2024, Corney et al., 2022). This growing prevalence represents not only a personal and emotional burden for patients and their families but also a major societal and economic challenge, as healthcare systems struggle to manage the long-term care needs of affected individuals. Alzheimer's is characterized by cognitive decline and memory loss, caused by structural alterations in brain regions, "including the hippocampus and prefrontal cortex" (Abouelmagd et al., 2024,Safiri et al., 2024). While Alzheimer's is prevalent and affecting millions of families, the diagnosis and treatment for it still lacks efficiency.

Currently, there exist multiple therapies for clinically diagnosed AD, but the onset of neurodegeneration often occurs years before diagnosis. According to Arafa et al., Associate Professor in the Department of Computer Engineering at Tanta University, the disease first begins with "a gradual deterioration of nerve cells, [then] advances to an acute stage of dementia that leaves patients unable to carry out fundamental daily activities" (Arafa et al., 2023). This late treatment leads to damage to brain cells that are vitally important and cause loss of "brain function, a decrease in mental skills, language problems, and a decrease in the ability to construct logical thoughts" (Arafa et al., 2023) The diagnosis of Alzheimer's typically occurs in two stages: when the patient has symptoms - which is often when the disease has already progressed significantly - and through neuroimaging techniques, but these

techniques are often invasive and unavailable to most patients. Early detection is crucial because the neurodegenerative processes underlying Alzheimer's begin years, sometimes decades, before clinical symptoms appear. Identifying risk factors or early biomarkers could allow interventions that slow disease progression, preserve cognitive function, and improve quality of life. The lack of non-invasive and effective diagnosis techniques highlights the need for new early detection methods.

Recently, there have been ongoing discoveries over the correlation between Adverse Childhood Experiences (ACEs) and the onset of Alzheimer's Disease. Adverse Childhood Effects include stressful and traumatic events, including "various forms of abuse (physical and emotional), neglect (physical and emotional), household dysfunction (such as substance misuse, incarceration, chronic depression or mental illness, domestic violence" and more, that have significant implications to the health of adolescents. According to Bethell et al., Professor at Johns Hopkins University Bloomberg School of Public Health, approximately one billion children, encompassing around 58% of the North American population aged between two and seventeen years have experienced such violence (Abouelmagd et al., 2024, Bethell et al., 2019). These ACEs affect the biological embedding of the brain, as majority of the structural and functional development to the brain, particularly the parts crucially involved in memory and cognitive function, occur during childhood development. According to Berens et al., Clinical Research Fellow at the Stanford University, "Dramatic growth in gray and white matter occurs in the first 2 years of life", encompassing nearly 80-90% of the adult-brain capacity. More specifically, Förster et al., junior professor at the University of Hamburg, explains that the hippocampus, involved in memory, learning, and stress regulation, is increasingly "susceptible to alterations during maturation" (Förster et al., 2021) In turn, typical neurodevelopment allows the brain to adapt to environmental factors and develop increasingly complex cognitive functions early in life, but this also leaves the brain

susceptible to negative environmental exposures (Berens et al., 2017). ACEs have been linked to “variation in brain structure and function”, affecting regions involved in learning, memory and general cognitive functioning - namely the hippocampus, amygdala and prefrontal cortex (PFC) (Berens et al., 2017). In particular, “hippocampal volume has repeatedly been shown to be negatively associated with childhood maltreatment”, explaining that the gray matter present in the hippocampus in adults with adverse childhood experiences is notably reduced compared to their healthy counterparts (Förster et al., 2021). Bigio et al., Research Assistant Professor at the NYU Grossman School of Medicine expands on this, citing that these structural changes are mediated by stress-related and metabolic pathways, more specifically alterations in mitochondrial function - critical for effective brain-cell communication - and insulin signaling. Insulin signaling "regulates neuroplasticity and is reflective of cerebral hypo-metabolism and aberrant intrinsic connectivity of intra-and inter-hippocampal circuits” (Bigio et al., 2023), or in other words, in the absence of insulin signaling, neurons don’t receive sufficient energy to function or maintain connections, resulting in damaged memory, learning, and overall brain communication. Corney et al., PhD Researcher at the IMPACT Institute, Australia, explore this more deeply, explaining that high glucocorticoid levels - a hormone released under extreme stress conditions - gradually increase neuron cell death. Neurons are essentially specialized brain cells that are involved in communication, cognitive functioning, learning, and memory formation. Damage to these cells results in smaller hippocampus volumes, as well as smaller PFC gray matter volumes, responsible for high-level cognitive functioning (Berens et al., 2017). Additionally, “ACEs have been associated with dysfunctional neurodevelopment, as well as diminished development of the hypothalamic pituitary adrenal (HPA) axis”, which is involved in the release of stress hormones (Corney et al., 2022). The hippocampus in particular is especially vulnerable to increased stress levels because of its high density of glucocorticoid receptors

(Förster et al., 2021). The structural and functional alterations in the brain resulting from increased stress levels are highly correlated with the increase of Alzheimer's. As explained by (Bigio et al., 2023), these disruptions in neuroplasticity and hippocampus functioning "is one of the steps in the irreversible activation of the cascade leading from mood disorders to AD", highlighting the potential connection between ACEs and the onset of Alzheimer's.

As explained previously, Alzheimer's Disease is a neurodegenerative disease primarily caused by the harmful accumulation of amyloid beta and tau proteins. As explained by Wu et al., Assistant Professor in Molecular Biology at UT Southwestern Medical Center, "AD is characterized by extracellular accumulation of amyloid beta-protein (Aβ) forming senile plaques, abnormal phosphorylation of tau protein, and neuronal loss" (Safiri et al., 2024, Wu et al., 2025). The clumps of amyloid-beta protein attach to neurons extracellularly and negatively impact their functioning. Additionally, the tangling of tau proteins, normally involved in stabilization of neurons, disrupts their internal structure. As a result, there is a cascade of neuron death. This is key in understanding the relationship between ACEs and Alzheimer's Disease, as the harmful effects of ACEs are directly related to the most prevalent causes of AD. According to Huang et al., Research Assistant Professor at the Tulane University, several experiments in laboratory animals have shown that the excess production of glucocorticoid, the stress hormone closely associated with ACEs, rapidly increases the formation of Aβ. Furthermore, "elevated glucocorticoid also exacerbates the development of neurofibrillary tangles [caused by tau proteins]" (Huang et al., 2023). This direct link between ACE effects and early formation of AD can be seen in studies as well. In a study consisting of 1123 participants, done by Schickedanz et al., Assistant Professor at UCLA School of Medicine, showed a direct correlation between the ACE score and the increased probability of developing dementia. These experiences included " emotional abuse, sexual abuse, exposure to intimate part- ner violence" and more "had the highest probability of a

positive screen for dementia" (Schickedanz et al., 2021). Additionally, Günak et al., Research Associate at University College London, performed a meta-analysis of studies with over 1.6 million participants combined and revealed that adults with PTSD had a 61% higher risk of developing dementia. These findings highlight that the biological effects of early-life stress, more specifically increased neuronal damage, are not just theoretical - they appear in real-world cognitive outcomes later in life. The strong associations between ACEs and dementia suggest that early adversity may accelerate the pathological processes associated with Alzheimer's Disease, making individuals more vulnerable to cognitive decline.

While existing research shows the clear links between ACEs and the increased risk of Alzheimer's Disease, there still remain significant gaps in this growing field of study. Many studies focus on mainly the biological mechanisms, conducting meta-analysis studies or performing MRI and PET scans. Additionally, the surveys in most recent studies did not take into account ethnic and educational diversity, and thus "future studies expanding various populations, and investigating the influence of rural and urban differences are needed" (Corney et al., 2022). Furthermore, much of the research was cross-sectional and relied on retrospective self-reporting of ACEs, a scale that was used differently throughout the studies. There is also limited work connecting specific types of early-life adversity - such as emotional neglect, physical abuse, or household dysfunction - to measurable biomarkers of AD. To address these gaps, the current study aims to examine how specific types of adverse childhood experiences are associated with biomarkers of Alzheimer's Disease in ethnically and educationally diverse populations.

Methods

The present study is a systematic review and meta-analytic approach to examine the differentiating associations between specific types of adverse childhood experiences and

clinical outcomes associated with Alzheimer's Disease. Rather than treating ACE exposure as a cumulative score of 10, this study focuses on separating ACE scores into different subtypes, namely physical abuse, emotional abuse, sexual abuse, physical neglect, emotional neglect, and household dysfunction, to determine which categories demonstrate the strongest associations with AD-related outcomes. This approach addresses a gap in the current body of knowledge, wherein most prior papers and analyses have researched ACEs as a cumulative measure rather than analyzing the validity of individual ACE categories and their relation to cognitive dysfunction (Corney et al., 2022; Abouelmagd et al., 2024).

Search Strategy

A literature search was conducted using the PubMed database to identify peer-reviewed studies examining the relationship between ACEs and Alzheimer's Disease, or related outcomes. The following search terms and Boolean combinations were used: ("adverse childhood experiences" OR "childhood trauma" OR "early life adversity" OR "childhood maltreatment") AND ("Alzheimer's disease" OR "dementia" OR "cognitive decline" OR "amyloid" OR "tau" OR "neurodegeneration"). The search was limited to studies published in English between January 2010 and December 2025, to reflect and analyze the significant growth in ACE-related research following the publication of the original Felitti et al. (1998) ACE study.

Inclusion and Exclusion Criteria

Studies were included if they met all of the following criteria: (1) the study involved human participants; (2) ACE exposure was measured using a defined and replicable instrument, such as the CDC-Kaiser ACE Questionnaire (About Adverse, 2026) or a valid equivalent; (3) the study reported outcomes related to Alzheimer's Disease, including but not limited to clinical dementia diagnosis, performance on validated cognitive assessments, such

as the SCD-Q Questionnaire (Rami et al., 2014), or measurable AD biomarkers such as amyloid-beta or tau protein levels (analyzed and identified through neuroimaging analysis); (4) the study reported a quantifiable effect size, such as an odds ratio (OR), relative risk (RR), or hazard ratio (HR), or provided sufficient data to calculate one; and (5) the study was published in a credible peer-reviewed journal. Studies were excluded if they involved only animal subjects, lacked a comparison or control group, used undefined or non-standardized ACE measurement, or if full-text access was unavailable.

Study Selection Process

The study selection process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). Following the initial database search, all duplicate papers were removed. Titles and abstracts of all remaining papers were screened against the inclusion and exclusion criteria mentioned above. Studies that clearly did not meet the criteria based on title and abstract alone were excluded at this stage. Full-text articles of all remaining studies were then read and assessed for final eligibility. A PRISMA flow diagram was constructed to document the number of studies identified, screened, assessed for eligibility, and ultimately included in the final analysis.

Data Extraction

Data were systematically extracted from each included study using an extraction software called Rayyan. The following variables were recorded for each study: author(s) and year of publication, study design (e.g., longitudinal cohort, case-control, cross-sectional), sample size, participant demographics including age, sex, location, race/ethnicity and educational background, ACE measurement instrument used, specific ACE subtypes measured, outcome measure used (e.g., dementia diagnosis, cognitive test score, biomarker level), and any notable limitations acknowledged by the authors.

Quality Assessment

The methodological quality of each included study was assessed using the Newcastle-Ottawa Scale (NOS), a widely used instrument for assessing the quality of non-randomized observational studies in meta-analytic papers. The NOS evaluates studies across three domains: selection of study groups (four items), comparability of groups (two items), and ascertainment of exposure or outcome (three items), resulting in a maximum score of nine stars. Studies scoring seven or above were classified as high quality, four to six as moderate quality, and below four as low quality.

Study Selection

The initial PubMed database search, conducted using Boolean combinations of ACE-related and Alzheimer's disease-related search terms, yielded 159 records. These articles were exported into the Rayyan software system to detect duplicates and further the screening process. At this stage, no duplicates were identified. All 159 records underwent title and abstract screening against the inclusion and exclusion criteria, mentioned in the previous section, and resulted in the exclusion of 107 studies. The primary reasons for exclusion at this stage were: (1) outcomes were unrelated to AD, dementia, or cognitive decline (e.g., studies focused on depression, psychosis or suicidal behaviour) and thus were unrelated to the present study; (2) review articles, editorials, or commentary pieces with no primary data; (3) animal studies; and (4) studies examining childhood adversity primarily in the context of psychiatric disorders without testing for cognitive or neurological outcomes. The remaining 52 records received full-text assessment. Of these, 28 were excluded: 13 reported only beta coefficients or correlation coefficients rather than the preferred odds ratios, hazard ratios, or relative risk values that would allow the distinguishing between different types of ACEs and their cognitive outcomes; 7 measured outcomes unleaded to AD or

dementia when reviewed; and 8 were excluded due to being reviews, animal studies, or using non-standardized ACE measurement instruments that were not detected in the earlier screening. A total of 24 studies were included in the final systematic review, the process of which is documented in the PRISMA flow diagram (Figure 1).

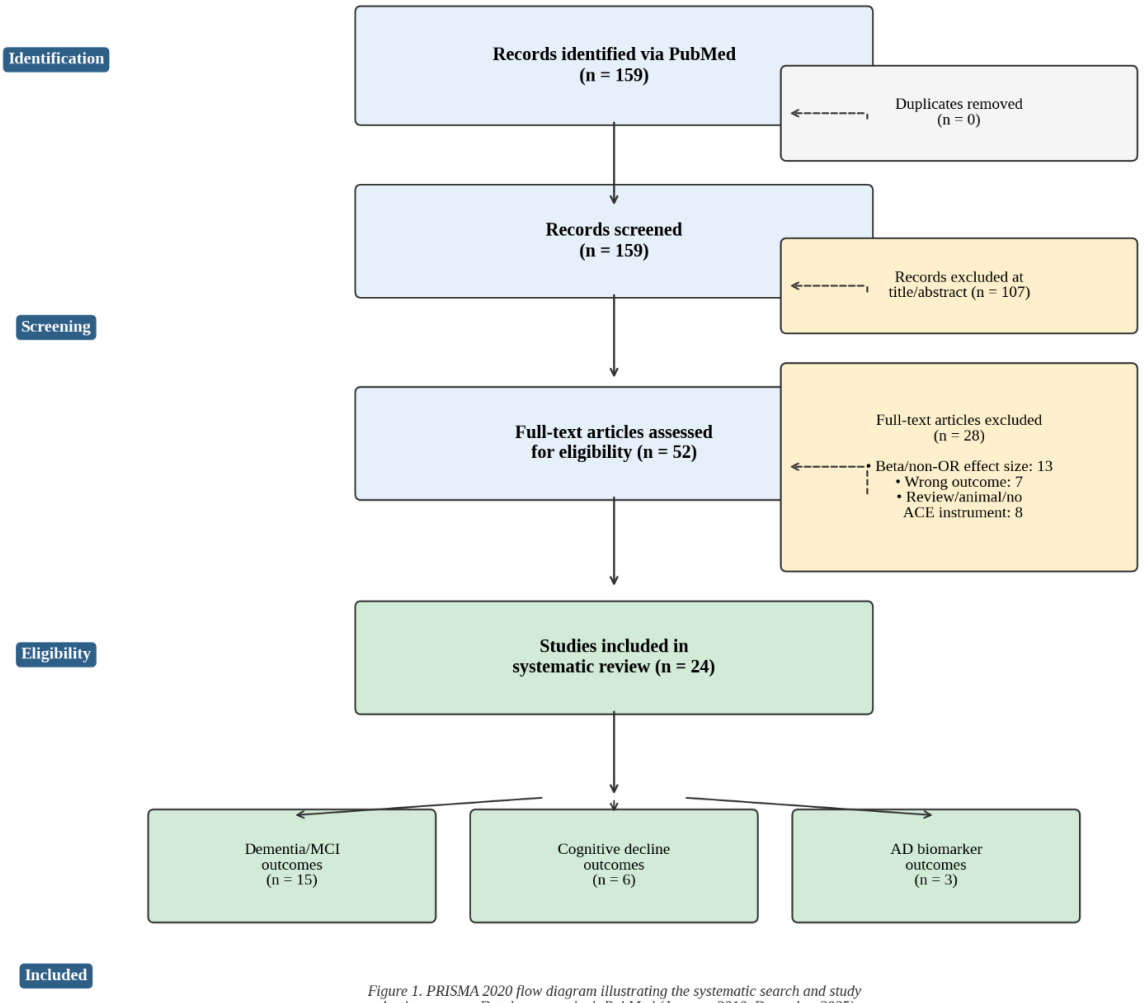

Figure 1. PRISMA 2020 flow diagram illustrating the systematic search and study selection process. Database searched: PubMed (January 2010–December 2025).

Characteristics of Included Studies

The 24 included studies were published between 2012 and 2026. Study designs included longitudinal prospective cohort studies (n=9), cross-sectional analysis of surveys (n=13), and one longitudinal clinical-pathologic cohort with brain autopsy outcome verification, which essentially means that the researchers did regular clinical check-ups on the patients, and later verified the cognitive decline outcome with post-death autopsy. Sample sizes ranged from 96 (Burri et al., 2013) to 434,215 (Günak et al., 2025). The studies also spanned diverse geographic locations, including the United States (n=14), China (n=4), the United Kingdom (n=2), Japan (n=1), Australia (n=2), and one multi-country setting. Outcome measures included clinical dementia diagnosis or screening (n=10), subjective cognitive decline (n=8), objective cognitive impairment (using a set of standardised tests) (n=4), and AD-specific biomarkers including amyloid PET and brain autopsy-confirmed cognitive decline (n=2). Full study characteristics are reported in Table 1.

Quality Assessment

Using the Newcastle-Ottawa Scale (Wells et al., 2000), 19 of the 24 studies were rated high quality (NOS ≥7) and 4 were rated moderate quality (NOS 4-6). Moderate-quality studies were primarily rated as such for non-standardized ACE instruments, namely Radford et al., 2019; Burri et al., 2013; and Le et al., 2025, or exclusive reliance on self-reported cognitive outcomes without clinical verification (Brown et al., 2022). The strongest studies had prospective designs with strong evidence: documented maltreatment records (Assuras et al., 2025; Maxfield et al., 2026) that avoids recall bias, exaggeration, etc and provides an accurate documentation of abuse; large-scale UK-biobank data to verify cognitive outcomes (Günak et al., 2025; Li et al., 2025); or, the most reliable, autopsy-confirmed cognitive outcomes (Wilson et al., 2012). NOS Scores are reported in full in Table 2.

Associations between ACE Subtypes and Alzheimer's Disease Outcomes

Across all 24 included studies, the overall direction of evidence consistently supported a positive association between adverse childhood experiences and an increased risk of cognitive decline.

Emotional Neglect

Emotional neglect was most consistently associated with the ACE subtype with AD-related neurological outcomes among the rest. Wilson et al. (2012) utilized brain-autopsy confirmed cerebral infarction - that is, physically examining the brain tissue after death to confirm cognitive decline - found that emotional neglect was the only ACE subtype significantly associated with chronic cerebral infarction (areas of the brain tissue decaying due to blocked blood vessels) with a 2.83-fold higher likelihood of infarction at high neglect levels compared to low levels. No other subtype measured - including parental intimidation, parental violence, family turmoil, or financial need - reached significance, showing that emotional neglect was by far the most essential in affecting cognitive decline. He and Yang (2024) used latent class analysis (a statistical method that groups people into categories based on patterns in their data) on 7,222 Chinese older adults and similarly found that a child maltreatment latent class (incorporating neglect alongside physical and emotional abuse) was significantly associated with cognitive impairment (OR = 1.37, 95% CI: 1.12 - 1.68), while a household dysfunction class was not (OR = 1.06, 95% CI: 0.90-1.26), providing a cross-cultural perspective on this subtype differentiation.

Physical and Sexual Abuse

Physical and sexual abuse were associated with cognitive impairment across multiple studies. Assuras et al. (2025) and Maxfield et al. (2026) used documented legal records of

childhood maltreatment and found that maltreatment was broadly associated with significant higher risk of cognitive impairment and amnestic CIND (Cognitive Impairment with No Dementia) in late midlife. In other words, people with documented childhood maltreatment were 86% more likely to show early-stage cognitive decline, and 68% more likely to show the specific memory-focused decline most associated with Alzheimer's progression. Terry at al. (2023) analyzed individual ACE categories from the BRFSS (Behavioral Risk Factor Surveillance System - a country-wide phone survey run by the Center of Disease Control and Prevention collecting data from hundreds of thousands of American adults) and found that emotional abuse and physical abuse were each independently significant, meaning those two specific ACE types on their own significantly increased the odds of SCD. Specifically, Chiow and Cicero (2025) found that childhood abuse was associated with 4.3 times higher odds of SCD in transgender and nonbinary adults.

Household dysfunction

Household dysfunction was one of the more inconsistent associations with AD-related outcomes compared to maltreatment-type ACEs, particularly when analyzed in isolation. As noted above, He and Yang (2024) found that household dysfunction was not associated with cognitive impairment. Additionally, Wilson et al. (2012) found that family turmoil did not predict cerebral infarction, while emotional neglect had significant associations. However, Terry et al. (2023) did identify that specific household dysfunction items, including family mental illness, substance abuse, and incarceration, each independently (rather than category as a whole) predicted SCD, suggesting that individual components in this category have significance. The reason this likely occurs is because the broad household dysfunction category average together high-risk components, like parental mental-illness, with low-risk

items, like parental divorce, resulting in a lower insignificant average despite individual categories having repercussions.

Cumulative ACE Exposure and Dose-Response Relationship

Studies that used ACE exposure as a cumulative score rather than dividing it into categories consistently found positive dose-response relationships with AD-related outcomes, meaning the more ACEs a person had, the higher their risk of cognitive impairment. Tani et al. (2020) found that ≥3 ACEs was associated with a 2.18-fold increased risk of dementia, and a 1.78 times higher risk when controlled for other factors like education, social relationships, and health behaviors. Günak et al. (2025) found that each additional ACE a person had was associated with a 10% higher dementia risk, particularly because ACEs increase PTSD symptoms (75% of the time), leading to higher dementia risk. Li et al. (2025) replicated this finding in both the UK Bioband and CHARLS cohorts, which significantly strengthens this finding. Chen et al. (2026) observed the effects when childhood and adulthood adversity co-occured, and found a 3.28 times higher dementia risk, suggesting that adversity accumulates throughout one's life, leading to more extreme implications.

Null and Inconsistent Findings

Two studies found no significant association between childhood adversity and cognitive outcomes. Lian et al. (2024) found no association between childhood domestic adversity and MCI (mild cognitive impairment) or dementia. Hayes-Larson et al. (2025), using racially diverse cohorts, found largely null associations with only a small effect on verbal memory decline, and no association with amyloid burden. Both studies attributed null findings to potential measurement limitations - their ACE instruments captured household dysfunction items more than direct maltreatment; selective survival bias - people who experienced severe childhood adversity and had worse health outcomes may have died before

reaching the age of enrollment in the study; or heterogeneity across racial and demographic subgroups - Hayes-Larson noted that prior research found potentially protective effects of adversity in Black Americans specifically, but these subgroup differences canceled each other out and resulted in null findings when Black, White, Asian, and Latino participants were pooled and analyzed together.

Discussion

This systematic review examined the associations between specific adverse childhood experience subtypes and Alzheimer's disease-related outcomes across 24 studies. The findings broadly support the hypothesis that ACE exposure in childhood is associated with increased risk of dementia and subjective cognitive decline in adulthood. However, the evidence suggests that this association is not uniform across all ACE subtypes; a distinction that has been largely overlooked in prior research treating ACEs as a single cumulative measure.

The most consistent finding across the included studies was the association between maltreatment-type ACEs - particularly emotional neglect - and AD-related neurological outcomes, in contrast to the weaker and less consistent associations observed for household dysfunction (Wilson et al., 2024; He and Yang., 2024). This subtype differentiation is meaningful because it suggests that the relationship between childhood adversity and dementia is not simply a function of cumulative stress exposure, but rather depends on the nature and frequency of the adversity experienced. Emotional neglect tends to be persistent and repeated rather than episodic; while most ACEs happen at specific moments and are episodic, like getting hit or witnessing a violent incident, emotional neglect is different because it's not one event - it's an ongoing absence. A child who is emotionally neglected experiences that neglect every day for years, throughout the most critical period of brain

development. This exerts sustained pressure on the developing stress response system, particularly the HPA axis and glucocorticoid production which, as explained previously, is especially vulnerable during childhood development. A single stressful event activates this system temporarily, but years of emotional neglect keeps it chronically activated, which over time causes the hippocampal damage and neurological changes associated with Alzheimer's Disease.

Household dysfunction, by contrast, demonstrated weaker and less consistent associated with AD outcomes, especially when analyzed separately from maltreatment-type ACEs, like emotional neglect or physical abuse. This pattern is particularly important because the original CDC-Kaiser ACE questionnaire combines household dysfunction and maltreatment items into a single cumulative score, which obscures meaningful differences in risk between these categories. A score of 3, for instance, could reflect experiences like emotional neglect, physical abuse, and sexual abuse, or it could include household-level adversities like parental divorce, household substance abuse, or familial incarceration - instances that reflect very different risk-levels despite having the same score. By collapsing everything into one number, the framework treats these as equivalent when they may not be.

Interestingly, the two null findings in this review - Lian et al. (2024) and Hayes-Larson et al. (2025) - provide important context rather than contradicting the overall pattern of evidence. Both studies measured adversity using instruments weighted toward household dysfunction and general childhood challenges rather than specific maltreatment experiences: Lien et al. (2024) used a domestic adversity scale focused on poverty and verbal abuse while Hayes-Larson et al. (2025) used a 7-item questionnaire that was built around household dysfunction items like parental divorce and job loss. It is important to note that all 7 items in this questionnaire are household-level events that happened around the child rather

than things done directly to the child. Neither instrument was particularly good at capturing direct maltreatment like emotional neglect or physical abuse, which has been identified as the subtypes most consistently associated with dementia. So, given the subtype-specific pattern identified across other included studies, it is plausible that these null findings reflect the limitation of measuring adversity broadly rather than a genuine absence of an ACE-dementia relationship. Heterogeneity in ACE measurement instruments therefore represents both a methodological limitation and a substantive finding: It's a methodological limitation because the fact that different studies measure ACEs differently makes it hard to compare findings across the field. However it is also a substantive finding because the pattern of which studies find significant results and which don't actually track with what kind of ACE instrument they used, suggesting that inconsistent findings are measuring fundamentally different categories under the same “ACE” label, and underscoring the importance of differentiating ACEs into subcategories for the most accurate and useful results.

Limitations

Several limitations of this paper should be acknowledged. First, the search was conducted using the PubMed database only, and while the search terms were broad, relevant studies from other databases, such as PsycINFO, Embase, or Scopus may have been missed. Second, the included studies used different ACE measurement instruments, including the CDC-Kaiser tool, Childhood Trauma Questionnaire, custom cohort-specific surveys, etc. This makes it difficult to draw precise conclusions about which ACE subtypes carry the greatest risk because it’s unclear whether the discrepancies were due to actual differences or because of different definitions of specific categories. Third, the majority of studies examining ACE subtypes in detail were conducted in predominantly White Western populations while findings from ethnically diverse samples viewed ACE scores cumulatively. This constrains

the generalizability of conclusions about subtype differentiation. Fourth, most included studies relied on adults self-reporting events of childhood adversity that occurred 50 - 60 years ago. This introduces recall bias as a consistent limitation across the articles as human memory is imperfect and people may misremember, underreport traumatic experiences, or interpret the same childhood event differently at different life stages. Finally, eight of the 24 included studies used subjective cognitive decline tests as an outcome measure in this review, which means that a substantial portion of the evidence relies on self-reported cognitive symptoms rather than clinically verified dementia diagnoses or neuroimaging scans showing amyloid builtup, the strength of the correlation discussed.

Implications

The findings of this review carry meaningful implications for early detection of AD, policy, and future research. Clinically, the consistent association between maltreatment-type ACEs and AD-related outcomes suggests that doctors, especially primary care physicians and geriatricians who work with older adults, should perform routine ACE screening with particular attention to emotional neglect and physical abuse. This could help identify individuals at elevated risk for cognitive decline decades before the onset of symptoms, and consequently help prevent the digression of the disease. Given that neurodegeneration begins years before initial clinical diagnosis, as discussed previously, early identification of high-risk individuals could help facilitate timely intervention, lifestyle modifications, and more frequent cognitive monitoring.

From a policy perspective, the dose-response relationships observed introduce the idea of upstream child maltreatment prevention as a potential dementia prevention strategy, or in other words, addressing the root cause of dementia - ACEs - before the disease actually develops, rather than treating dementia after diagnosis, could reduce the number of

Alzheimer's affected patients and could work as a prevention strategy. In practice, this would include improving systems that identify and report child abuse, training teachers and pediatricians to recognize and respond to trauma, and educating parents on the dangers of ACEs and proper healthy parenting skills. This would yield long-term public health benefits extending beyond childhood outcomes to include reduced dementia burden, both psycholocal and financial, in populations.

Future Directions

For future research, this review highlights the urgent need for studies that measure and analyze ACE subtypes independently rather than as cumulative scores to see the impact specific types of neglect have on cognitive impairment, ultimately leading to better prevention measures. Additionally, longitudinal designed studies with clinically verified cognitive outcomes, using methods like brain-imaging techniques, would yield more accurate results than the current cross-sectional ones, as it would remove limitations like recall-bias and provide stronger evidence for causal correlation between ACE subtypes and the onset of Alzheimer's disease. Finally, it is evident that there is a relationship between ACEs and neurological degeneration, but the pathway to this degeneration remains unclear. So, there is a growing need for studies to examine the biological pathways, particularly glucocorticoid dysregulation and hippocampal structural changes, through which specific ACE types may accelerate AD-related outcomes.

Conclusion

This systematic review provides evidence that adverse childhood experiences are associated with increased risk of Alzheimer's disease and related cognitive decline across diverse populations and study designs. Most importantly, the findings suggest that not all ACE subtypes result in equal risk: maltreatment-type ACEs, particularly emotional neglect

and physical abuse, demonstrated stronger and more consistent associations with AD-related outcomes when compared to household dysfunction-type ACEs. Additionally, the two null findings in this review support this interpretation, as both studies primarily measured household-level adversity rather than direct maltreatment, resulting in no significant relationship being found. These patterns suggest that the current reliance on cumulative ACE scores may obscure meaningful subtype-level differences in dementia risk, and that future research adopting a more segregated approach to ACE classification will be essential in advancing the understanding of early-life origins of cognitive decline, eventually serving as much needed early-prevention treatments for Alzheimer's Disease.

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Appendix

Table 1. Characteristics of Included Studies (Data Extraction)

Note. ACE = Adverse Childhood Experience; OR = Odds Ratio; HR = Hazard Ratio; RR = Relative Risk; AOR = Adjusted Odds Ratio; CI = Confidence Interval; SCD = Subjective Cognitive Decline; CIND = Cognitive Impairment with No Dementia; MCI = Mild Cognitive Impairment; NOS = Newcastle-Ottawa Scale; NS = Not Significant; LCA = Latent Class Analysis; EHR = Electronic Health Record. ***Additionally, it should be noted that Assuras et al. (2025) and Maxfield et al. (2026) draw from the same prospective cohort and should be interpreted as converging rather than independent evidence.***

Author & Year	Study Design	Sample Size	Population	ACE Instrument	ACE Subtypes Measured	Outcome Measure	Effect Size	95% CI	Key Finding	NOS	Limitations
Assuras et al. (2025)	Prospective cohort	447	US, mean age 59.4 yrs	Documented court cases	Physical abuse, sexual abuse, neglect	Cognitive impairment (CIND); neuropsychological battery	AOR = 1.86 (CIND); AOR = 1.68 (amnestic CIND)	1.35–2.57; 1.20–2.34	Documented maltreatment associated with higher risk of CIND and amnestic CIND in late midlife	9	Documented cases only; small final sample
Brown et al. (2022)	Cross-sectional	National sample	US adults, BRFSS 2020	CDC-Kaiser ACE Questionnaire	Cumulative score (≥2 ACEs)	Subjective cognitive decline (SCD)	AOR = 1.69	1.36–2.10	≥2 ACEs significantly increased odds of SCD	6	Cross-sectional; self-reported SCD
Brown et al. (2025)	Cross-sectional	38,600	US adults ≥45 yrs, BRFSS 2023	CDC-Kaiser ACE Questionnaire	Cumulative score	SCD; depression as mediator	B = 0.066 (direct); B = 0.029 (mediated)	p < 0.001	ACEs associated with SCD; depression partially mediates	7	Cross-sectional; beta coefficients
Burri et al. (2013)	Cross-sectional	96	Former indentured Swiss child laborers	Structured interview; MMSE	Childhood vs adulthood trauma; PTSD	Cognitive function (MMSE, dementia screen)	PTSD+ worse on all cognitive tasks	Not reported as OR	PTSD symptoms associated with poorer cognitive function	5	Very small sample; no standard ACE instrument
Chen et al. (2026)	Population-based cohort	11,601	Chinese adults ≥45 yrs, CHARLS	CHARLS structured questionnaire	Cumulative ACE + AAE score	Incident dementia; stroke	HR = 1.11 (ACEs); HR = 3.28 (ACEs+AAEs)	1.05–1.18; 1.54–7.02	ACEs independently associated with dementia; combined ACE+AAE exposure 3.28× higher risk	8	Self-reported ACEs; short follow-up
Chiow & Cicero (2025)	Cross-sectional	206	US transgender/nonbinary adults ≥45 yrs	CDC-Kaiser ACE Questionnaire	Childhood abuse; household dysfunction; cumulative	SCD	aOR = 1.4 per ACE; aOR = 4.3 (abuse); aOR = 4.7 (dysfunction)	1.2–1.6; 1.88–10.02; 2.00–11.07	ACEs strongly associated with SCD in TGN adults	7	Small sample; cross-sectional
Cockell et al. (2025)	Longitudinal cohort	16,509	US adults, HRS 2008–2018	HRS childhood questionnaire (31 items)	31 childhood exposures before age 16	Incident cognitive impairment; dementia	RR = 1.49 (childhood depression); RR = 2.42 (poor health)	1.20–1.84; 1.62–3.61	Multiple childhood conditions associated with incident dementia	8	Not standard ACE instrument; retrospective
Günak et al. (2025)	Prospective cohort	434,215	UK Biobank, mean age 56.58 yrs	UK Biobank ACE questionnaire	Cumulative ACE score	Incident dementia (EHR-linked)	HR = 1.10 per ACE point	1.02–1.20	ACEs associated with dementia; PTSD mediated 75% of excess risk	9	Self-reported ACEs; UK Biobank not fully representative
Hayes-Larson et al. (2025)	Longitudinal cohort	2,447 (cognition); 331 (PET)	Racially diverse US older adults, KHANDLE+STAR	7-item adversity factor score (CFA)	Household dysfunction items	Cognitive decline (SENAS); MRI; amyloid PET	β = −0.013 SD/yr (memory); PR = 0.94 (amyloid)	−0.025 to −0.001; 0.79–1.12	Slightly faster verbal memory decline; no amyloid association	9	Household dysfunction only; no abuse/neglect items
He & Yang (2024)	Retrospective cohort	7,222	Chinese adults ≥60 yrs, CHARLS	CHARLS ACE questionnaire (11 types)	Child maltreatment vs household dysfunction (LCA)	Cognitive impairment (MMSE)	OR = 1.37 (maltreatment); OR = 1.06 (dysfunction, NS)	1.12–1.68; 0.90–1.26	Maltreatment associated with cognitive impairment; dysfunction was not	7	Retrospective; cross-sectional cognitive measure

Joshi & Yeo (2025)	Cross-sectional (SEM)	35,754	US adults ≥45 yrs, BRFSS 2023	CDC-Kaiser ACE Questionnaire	Cumulative score	SCD	β = 0.136	p < 0.001	ACEs associated with SCD; social support attenuated relationship	7	Cross-sectional; beta coefficient
Le et al. (2025)	Cross-sectional	6,693	Indonesian adults ≥50 yrs	IFLS childhood cluster measures	Unhealthy childhood health; poor SES	Dementia risk (clinician-assessed)	OR = 1.17 (health); OR = 1.39 (SES)	1.00–1.38; 1.15–1.68	Poor childhood health and SES independently associated with dementia	6	Not standard ACE instrument; borderline significance
Li et al. (2025)	Longitudinal cohort	147,942	UK Biobank + CHARLS replication	UK Biobank ACE questionnaire	Emotional neglect, physical/emotional abuse, sexual abuse, physical neglect	Incident dementia; Parkinson's disease	HR = 1.14 per ACE	1.08–1.21	Greater ACE exposure associated with dementia and PD; amplified by genetic risk	9	Self-reported ACEs; retrospective
Lian et al. (2024)	Cross-sectional	1,568	Australian adults 72–79 yrs, PATH	17-item childhood adversity scale	Cumulative; ≥3 adversities; LCA	MCI or dementia (clinically validated)	No significant association	NS	No significant association found — null finding	7	17-item scale not standard ACE instrument
Maxfield et al. (2026)	Prospective cohort (5 waves)	447	US, documented maltreatment cases	Documented court cases	Physical abuse, sexual abuse, neglect	Cognitive functioning; AD8 dementia screen	Poorer performance on 11/12 cognitive tasks; steeper decline	Cohen's d 0.28–0.42	Maltreatment predicted steeper cognitive decline into late midlife	9	Same cohort as Assuras et al.; attrition over 5 waves
Radford et al. (2019)	Cross-sectional	336	Aboriginal Australians ≥60 yrs	Childhood trauma (logistic regression item)	Childhood trauma (yes/no)	All-cause dementia; Alzheimer's disease	Independent predictor of dementia	OR not reported; p < 0.05	Childhood trauma independently predicted dementia in Aboriginal Australians	6	Small sample; no validated ACE scale
Ren et al. (2023)	Longitudinal cohort	15,282	US older adults, HRS 2010–2020	HRS ACE questionnaire (6 ACEs)	Cumulative ACEs (0, 1, ≥2)	Incident dementia; Alzheimer's disease specifically	HR = 1.87 (dementia); HR = 1.85 (AD)	1.45–2.42; 1.22–2.80	ACEs associated with dementia and AD in adults with sustained obesity	8	Effect only in obese subgroup; self-reported ACEs
Roberts et al. (2022)	Cross-sectional	1,755	Former NFL players, mean age 57.2 yrs	Standard ACE Questionnaire	Cumulative ACEs (≥4 vs 0)	Dementia symptoms (AD8 screen)	RR = 1.48	1.22–1.79	≥4 ACEs associated with 48% greater risk of positive dementia screen	7	Specialized population; concussion history as confounder
Schickedanz et al. (2021)	Cross-sectional	1,223	US adults ≥65 yrs, PSID 2017	PSID Childhood Retrospective Survey	Cumulative score (≥4 vs 0)	Dementia screen (AD8)	aOR = 1.79	1.05–3.04	≥4 ACEs associated with 79% higher odds of positive dementia screen	7	Cross-sectional; screening only, not clinical diagnosis
Tampubolon & Li (2025)	Cross-sectional (HCAP)	12,862	England, Europe, China, US; mean age 69 yrs	Latent construction from retrospective childhood data	Cumulative early life adversity (latent variable)	MCI; dementia (multinomial logistic regression)	RR = 1.50 (MCI); RR = 1.52 (dementia)	1.16–1.92; 1.12–2.06	Early life adversity associated with MCI/dementia in England/Europe but not China/US	7	Preprint; retrospective ACE; selective mortality bias
Tani et al. (2020)	Prospective cohort	17,412	Japanese adults ≥65 yrs, JAGES	Survey-based ACE questionnaire (7 types)	7 ACE types; cumulative (0,1,2,≥3)	Incident dementia (care insurance system)	HR = 2.18 (≥3 ACEs); HR = 1.78 (adjusted)	1.42–3.35; 1.15–2.75	≥3 ACEs associated with 2.18× higher dementia risk in Japanese older adults	9	Short 3-year follow-up; care insurance dementia definition
Terry et al. (2023)	Cross-sectional	National sample	US adults, BRFSS 2019–2020	CDC-Kaiser ACE Questionnaire (8 categories)	8 individual ACE categories	SCD (memory loss)	Multiple individual ACE ORs; all p < 0.05	Various	Individual ACE categories differentially associated with SCD	7	Cross-sectional; self-reported SCD
Voyer et al. (2023)	Cross-sectional	National sample	US adults ≥45 yrs, BRFSS 2020	CDC-Kaiser ACE Questionnaire	Cumulative ACEs (≥2)	SCD	AOR = 1.69	1.36–2.10	≥2 ACEs significantly associated with SCD	7	Cross-sectional; self-reported SCD; overlaps

											with Brown et al. 2022
Wilson et al. (2012)	Longitudinal clinical-pathologic	192 (autopsy)	US older adults, Rush MAP; mean age at death 88.5 yrs	16-item CTQ-adapted scale (α=0.84)	Emotional neglect, parental intimidation, parental violence, family turmoil, financial need	Chronic cerebral infarction (brain autopsy)	OR = 1.097 (emotional neglect); OR = 2.83 (high vs low neglect)	1.048–1.148; 1.95–4.09	Only emotional neglect (not other subtypes) significantly associated with cerebral infarction	9	Small autopsy sample; retrospective ACE recall

Table 2. Newcastle-Ottawa Scale Quality Assessment

Note. ★ = 1 star awarded; ★★ = 2 stars awarded (comparability domain only); — = criterion not met or not applicable. Quality ratings: High = 7–9 stars; Moderate = 4–6 stars; Low = 0–3 stars. Selection domain maximum = 4 stars; Comparability domain maximum = 2 stars; Outcome/Exposure domain maximum = 3 stars.

Author & Year	Study Design	Representativeness(★)	Non-exposedselection (★)	ACEascertainment (★)	Outcome absentat start (★)	Comparability(★★)	Outcomeassessment (★)	Follow-up length (★)	Follow-upadequacy (★)	Total(/9)	QualityRating
Assuras et al. (2025)	Prospective cohort	★	★	★★	★	★★	★★★	★	★	9	**High**
Brown et al. (2022)	Cross-sectional	★★	★	★	—	★	★	—	★	6	**Moderate**
Brown et al. (2025)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**
Burri et al. (2013)	Cross-sectional	★	★	★	—	★	★	—	★	5	**Moderate**
Chen et al. (2026)	Population-based cohort	★★	★	★	★	★★	★	★	★	8	**High**
Chiow & Cicero (2025)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**
Cockell et al. (2025)	Longitudinal cohort	★★	★	★	★	★★	★	★	★	8	**High**
Günak et al. (2025)	Prospective cohort	★★	★	★★	★	★★	★	★	★	9	**High**
Hayes-Larson et al. (2025)	Longitudinal cohort	★★	★	★★	★	★★	★	★	★	9	**High**
He & Yang (2024)	Retrospective cohort	★★	★	★	★	★★	★	—	★	7	**High**
Joshi & Yeo (2025)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**
Le et al. (2025)	Cross-sectional	★★	—	★	—	★★	★	—	★	6	**Moderate**
Li et al. (2025)	Longitudinal cohort	★★	★	★★	★	★★	★	★	★	9	**High**
Lian et al. (2024)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**
Maxfield et al. (2026)	Prospective cohort	★	★	★★	★	★★	★★★	★	★	9	**High**
Radford et al. (2019)	Cross-sectional	★★	—	★	—	★★	★	—	★	6	**Moderate**
Ren et al. (2023)	Longitudinal cohort	★★	★	★	★	★★	★	★	★	8	**High**
Roberts et al. (2022)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**

Author & Year	Study Design	Representativeness(★)	Non-exposedselection (★)	ACEascertainment (★)	Outcome absentat start (★)	Comparability(★★)	Outcomeassessment (★)	Follow-up length (★)	Follow-upadequacy (★)	Total(/9)	QualityRating
Schickedanz et al. (2021)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**
Tampubolon & Li (2025)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**
Tani et al. (2020)	Prospective cohort	★★	★	★★	★	★★	★	★	★	9	**High**
Terry et al. (2023)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**
Voyer et al. (2023)	Cross-sectional	★★	★	★	—	★★	★	—	★	7	**High**
Wilson et al. (2012)	Longitudinal clinical-pathologic	★	★	★★	★	★★	★★★	★	★	9	**High**